CREDIT PRE-SCREENING ENGINE: LOAN INTAKE OPTIMISATION

Individual Contribution Report

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Abstract

This report documents my contribution to the MUC02 Credit Pre-Screening Engine, focusing on CBIM coordination and Steps 5.1–5.2: dataset loading and inspection, data-quality assessment, cleaning decisions, and descriptive analysis of historical approval patterns. The dataset contained 12,000 loan applications and 19 original columns. Historical decisions comprised 6,353 approvals (52.9%) and 5,647 declines (47.1%). The review identified 441 missing monthly-income values, 1,039 missing credit scores associated with zero credit-history months, and 12 unusually high salaried-income values requiring unit correction. Approval rates varied substantially across credit-score bands and employment categories. These findings established the data-quality baseline and business context for the team's subsequent feature engineering and modelling.

I. Introduction and Assigned Responsibility

The MUC02 use case examined whether historical personal-loan application data could support a consistent preliminary screening workflow while retaining human oversight. My assigned responsibility was to coordinate the CBIM problem-framing work and complete Steps 5.1–5.2. This included understanding the business problem, identifying the dataset as the source of truth, inspecting the target and input fields, documenting data-quality issues, and analysing historical approval rates by credit-score band and employment type.

II. CBIM Problem Canvas

The business context described approximately 3,000 personal-loan applications per month, with about 25 minutes of manual review per application and nearly half of historical applications declined. The central question was which applications, if any, could be routed to a conservative auto-decline lane and at what model threshold. The dataset, MUC02_Credit_PreScreening_Dataset.csv, contained 12,000 applications and 19 original columns. The target field, decision, records past analyst judgement rather than actual loan repayment.

The success criterion was framed around a maximum 5% wrong-decline share: among applications placed in the auto-decline lane, no more than 5% should be applications historically approved by the credit team. The threshold should be selected on validation data and evaluated separately on held-out test data. The primary stakeholder identified in the team report was the Credit Policy Analyst, alongside credit analysts and affected applicants.

III. Dataset Inspection and Data-Quality Review

The dataset contained 12,000 rows and 19 original columns. The inspection found no duplicate rows or duplicate application IDs. The target distribution was 6,353 Approved applications (52.9%) and 5,647 Declined applications (47.1%), indicating a reasonably balanced historical decision outcome.

Data-quality issue	Count	Treatment / interpretation
Missing monthly income	441	Imputed using the median monthly income for the same employment type.
Missing credit score	1,039 (8.7%)	Retained and flagged as new-to-credit (is_ntc = 1); all had credit_history_months = 0.
Unusually high salaried income	12	Divided by 12 after review because the values were likely annual income entered in a monthly-income field.

Retaining the 1,039 no-score applications was important because every one had zero credit-history months. The team report records a 53.9% historical approval rate for this group. Therefore, a missing credit score should not be interpreted as an automatic decline reason; these applicants should remain eligible for human assessment.

IV. Historical Approval Pattern Analysis

The analysis compared historical approval rates across credit-score bands. Approval was 2.0% below 600 and 3.2% for 600–649, then increased to 32.5% for 650–699. It continued to rise to 54.4% for 700–749, 77.2% for 750–799, and 85.2% for scores of 800 or higher. Applicants with no score had a 53.9% approval rate.

Credit-score band	Historical approval rate
Below 600	2.0%
600–649	3.2%
650–699	32.5%
700–749	54.4%
750–799	77.2%
800+	85.2%
No score	53.9%

Employment-type analysis showed a 58.4% historical approval rate for Salaried applicants, 51.2% for Business Owners, 50.6% for Self-Employed Professionals, and 34.9% for Gig / Contract applicants.

Employment type	Historical approval rate
Salaried	58.4%
Business Owner	51.2%
Self-Employed Professional	50.6%
Gig / Contract	34.9%

The strongest descriptive change occurs between the 600–649 and 650–699 credit-score bands. However, these figures describe historical analyst decisions; they do not prove that a particular score or employment category causes default or justifies automatic rejection. In particular, the 53.9% approval rate for applicants with no score supports retaining NTC applicants for individual assessment.

V. Methodological Considerations and Handoff

A methodological limitation was identified in the current preprocessing workflow. Employment-specific median income values were calculated using the complete dataset before splitting the data. This creates a potential data-leakage risk because validation and test information influences imputation. For a leakage-safe evaluation, median values must be calculated using training data only and then applied to validation and test datasets. The model and downstream evaluation results should be regenerated after this correction.

My data-quality log and historical approval summaries were handed over to the team members responsible for affordability feature engineering, Logistic Regression modelling, threshold selection, business-impact estimation, and fairness analysis. The findings provided descriptive context, not final credit-policy rules.

VI. Conclusion

My contribution established the dataset baseline, documented the principal data-quality decisions, and identified historical approval patterns by credit score and employment type. The analysis highlighted the importance of preserving new-to-credit applicants and treating score and employment patterns as descriptive evidence rather than automatic lending rules. Since the target captures past analyst decisions rather than repayment outcomes, further validation using repayment data, training-only preprocessing, out-of-time evaluation, and ongoing fairness monitoring is necessary before any real-world deployment.